

Momentum v2 — Planning & Design Doc

Sep 24, 2026 · @Sherbaz

Overview

Momentum helps you run your life like an operation. It is an AI chief of staff: it interviews you about your goals (career, health, money, learning) and your fixed commitments, agrees a plan with you, builds each day around your real schedule, reminds you before every block, and re-plans when life happens. The AI proposes every change; you approve it.

Project	Momentum v2: run your life like an operation
Team	Sherbaz Riasat (solo: planning, design, development)
Vision	Everyone gets a chief of staff that turns "I want to..." into "here is what I do today", and keeps them on it
Status	MVP built and tested (11 automated tests), code on GitHub, deploying on Vercel
Presentation	Momentum Keynote (13 interactive slides)
MVP deadline	September 29, 2026

Problem & users

Ambitious people don't lack goals. They lack a system, and someone to answer to. Career, health, money and learning goals compete for the same few hours, none of them get scheduled, and by week three nobody is checking.

- **What problem are we solving?** Turning big, competing life goals into a realistic day-by-day plan that fits around work, classes and family, and staying consistent with it.
- **Who has it?** Students, young professionals and anyone building something on the side.

- **Why is it worth solving?** Most goals fail on execution, not ambition. Planners assume you already know the steps, habit apps don't know the bigger goal, and chatbots forget and change things silently.
- **What would a useful solution do?** Learn your life, agree a plan with you, tell you what to do today, remind you, and re-plan when you slip.

User	Situation	What they want
Student	Classes, clubs, career prep	A path from "internship by summer" to what to do on Tuesday
Young professional	Job, gym, savings, family	Promotion, fitness and savings goals that don't all collapse by March
Builder	Day job plus a side project	A plan that fits the 10 hours a week left after work

Requirements

Functional (all built in v2)

1. **Interview:** 11 questions capture name, program or job, fixed commitments (classes, work, gym, family), sleep, peak energy, weekly hours and 1–5 goals with a why; optional .ics calendar import; an editable summary before planning.
2. **Plan proposal:** Gemini (or the offline planner) proposes milestones and weekly tasks that fit the weekly hours and lighten busy weeks.
3. **Negotiation:** the user pushes back in plain English; changes come back as a diff (add / change / remove, with a reason each) to approve or reject; **Agree & lock** creates a numbered plan version.
4. **Daily schedule:** a deterministic scheduler places tasks around fixed commitments, with P1 / P2 / P3 tiers, high-energy work in peak hours, a 90-minute focus cap and a daily budget.
5. **Today:** Start / Done / Skip per block, carried-over work, morning brief in the chosen coach style.
6. **Reminders:** a notification before each block and a nudge 15 minutes in.
7. **Evening review:** done / partial / skipped, mood and energy; unfinished work becomes a re-plan proposal.

8. **Command Center:** goal health, risk feed with Fix, semester timeline, execution trend, workload heatmap, ontology graph, action log.
9. **Chief-of-staff chat:** questions ("what should I do now?") and change requests ("I'm sick until Wednesday") that return proposals.
10. **Week view, settings, backup export/import, demo mode.**

Non-functional

- Works on phones (responsive, installable PWA, offline shell).
- Private by default: data stays on the device in demo mode; the Gemini key stays on the server.
- Reliable AI: every response is validated (Zod); on failure the offline engine takes over.
- Tested: 11 automated tests for the scheduler, interview parser and planner.
- Accessible: keyboard navigation, visible focus, reduced-motion support.

User stories & acceptance criteria

#	User story	Acceptance criteria
US1	As a user, I want to tell Momentum about my life and goals, so that the plan fits my real schedule.	11-question interview; .ics import works; every parsed field (commitments, hours, goals) is editable before planning
US2	As a user, I want my goals broken into milestones and weekly tasks, so that I know exactly what to do.	Each goal gets dated milestones with a definition of done; weekly load never exceeds my hours; works without an AI key
US3	As a user, I want to negotiate the plan, so that I agree with it before it runs my days.	A request returns a diff with a reason per line; Approve applies it, Reject changes nothing; Agree & lock creates plan v1, later approvals create v2, v3...
US4	As a busy person, I want each day scheduled around my commitments, so that I don't have to plan it myself.	No block overlaps a fixed commitment; P1 first in peak hours; focus blocks ≤ 90 min; daily budget respected; locked blocks never move

#	User story	Acceptance criteria
US5	As a procrastinator, I want reminders and a coach, so that I actually start.	Reminder before each block with Start / Snooze / Skip; nudge 15 min in; morning brief in my chosen tone
US6	As a user, I want missed work re-planned, so that one bad day doesn't break the plan.	Evening review turns unfinished P1/P2 work into a proposal for the next days; optional work drops first
US7	As a user, I want to see if I'm on track, so that I can fix problems early.	Command Center shows on track / at risk / off track per goal; each risk has a Fix that drafts a proposal

MVP scope

The smallest useful version: one person, several life goals, an agreed plan, a scheduled day and a chief of staff that keeps them on it.

Priority	Features	Status
Must have	Interview, plan proposal, negotiation with diffs, Agree & lock, daily scheduler, Today view, reminders, evening review	Built
Should have	Command Center, chief-of-staff chat, week view, Gemini integration with offline fallback, PWA, backup export/import	Built
Nice to have	Ontology graph, animated landing page, interactive keynote	Built
Next (Phase 2)	Accounts and cloud sync (Supabase), Web Push when the app is closed, Google Calendar sync	Designed, not built

Not in this MVP: native store apps, payments, friend/partner features, team or coach editions.

Constraints & dependencies

- **External API:** Google Gemini (free key from Google AI Studio), called only from the server route `/api/ai`. Without a key, the offline planner and coach are used.
- **Hosting:** Vercel (free Hobby plan), deployed from GitHub; root directory `v2`.
- **Data:** stored in the browser in the MVP; Supabase schema (14 tables with row-level security) is written for Phase 2.
- **Privacy/security:** personal goals stay on the device; the API key never reaches the browser; AI requests are rate-limited.
- **Reminders:** fire while the app is open; closed-app reminders need Web Push (Phase 2).
- **Hardware:** none; any modern phone or laptop browser.

Design: user flow & screens

```
flowchart TD
  A[Landing page] --> B[Interview:<br/>11 questions]
  B --> C[Review & edit summary]
  C --> D[AI proposes plan]
  D --> E{Agree?}
  E -- Push back --> F[Diff: approve / reject]
  F --> E
  E -- Agree & lock --> G[Today]
  G --> H[Reminders + Start/Done/Skip]
  H --> I[Evening review]
  I --> J[Re-plan proposal]
  J --> G
```

A new user goes through the interview once, agrees a plan, and from then on lives in the daily loop: Today, reminders, evening review, re-plan.

Screen	Route	Purpose
Landing	<code>/</code>	Animated "run your life like an operation" page, live demo button
Interview	<code>/app/onboarding</code>	11 questions, .ics import, editable summary
Plan	<code>/app/plan</code>	Timeline, weekly load, negotiation chat, Agree & lock, versions

Screen	Route	Purpose
Today	<code>/app/today</code>	Morning brief, time-blocked day, P1/P2/P3, carried-over work
Week	<code>/app/week</code>	7-day calendar around fixed commitments
Review	<code>/app/review</code>	Evening review, re-plan proposal
Command Center	<code>/app/command</code>	Goal health, risks + Fix, timeline, trend, heatmap, ontology, action log
Settings	<code>/app/settings</code>	Coach style, reminders, AI status, backup, reset

Wireframes: [Momentum Wireframes](#). High-fidelity design: the v2 app itself (dark operations-console look, Geist typefaces, one blue accent, status colours for on track / at risk / off track). Presentation: [Momentum Keynote](#).

Data model & architecture

Everything is one typed ontology shared by the UI, the AI and the database: a person has goals, goals have milestones, milestones have tasks, and tasks become time blocks around fixed commitments.

Object	Key fields
Profile	name, program/job, wake, sleep, peak energy, weekly hours, active days, coach style, reminder lead
Course / FixedEvent	title, kind (class, work, gym, family...), weekdays, start, end
Goal	title, why, category (career, health, money, learning...), target date, success metric, priority
Milestone	goal, title, due, definition of done, status
Task	goal, milestone, title, tier (P1/P2/P3), estimate, energy, earliest, due, status
PlanVersion	version, summary, agreed at, task and milestone counts

Object	Key fields
Proposal	kind (initial, chat edit, re-plan), rationale, diff, status
TimeBlock	date, start, end, kind, task, state (pending, started, done, skipped, missed)
CheckIn / ActionLog / ChatMessage	reviews, every AI and user change, conversation

```
flowchart LR
  U[User<br/>phone · laptop] --> APP[Next.js app<br/>React + Zustand]
  APP --> AI[/api/ai<br/>Gemini + Zod/]
  APP --> SCH[Scheduler<br/>deterministic]
  APP --> ST[(Browser storage<br/>→ Supabase)]
  APP --> RM[Reminders<br/>Notifications]
  APP --> CAL[.ics import]
```

AI at the edges, a deterministic core: Gemini handles the interview, the plan, negotiation and coaching; tested TypeScript code places every block.

How components communicate

- GET /api/ai tells the app whether Gemini is configured; POST /api/ai runs one agent: intake , plan , revise or coach .
- Each agent returns JSON that is validated with Zod and retried once; if it still fails, the offline engine answers.
- The scheduler buildDay() runs in the browser whenever a day is built or rebuilt.
- Proposals are applied only after the user approves them, and every change is written to the action log.

Tech stack & version control

Layer	Choice	Why
Framework	Next.js 15 (App Router) + TypeScript	One codebase for pages and the server AI route
UI	Tailwind CSS v4, Geist fonts, Motion for animation	Fast to build a consistent, animated UI

Layer	Choice	Why
State	Zustand, saved in the browser	Works offline; maps 1:1 to database tables later
AI	Google Gemini 2.5 Flash via <code>@google/genai</code> , validated with Zod	Free tier, JSON output; key stays on the server
Scheduling	Custom TypeScript scheduler	Deterministic and unit-tested; no AI calendar math
Testing	Vitest (11 tests), Playwright end-to-end checks	Scheduler, interview parser and planner are covered
Mobile	PWA manifest + service worker	Add to Home Screen, offline shell, notification buttons
Hosting	Vercel (Hobby), root directory <code>v2</code>	Free, deploys on every push
Database (Phase 2)	Supabase Postgres + Auth, row-level security	Schema already written in <code>supabase/migrations</code>
Version control	Git + GitHub (<code>riyasattechnologies-arch/Momentum</code>), GitHub Desktop	<code>main</code> is the deployed version

Tasks & timeline

Date	Work	Status
Sep 24	Idea, planning doc, wireframes, v1 prototype	Done
Sep 25	v2 build prompt; v2 app: interview, plan negotiation, scheduler, Today, Week, Review, Command Center, chat	Done
Sep 25	Animated landing page, Supabase schema, roadmap, tests	Done
Sep 26	Interactive keynote reframed to "run your life like an operation"	Done
Sep 26–29	Push to GitHub, deploy on Vercel	Done / in progress

Date	Work	Status
Sep 29	Add Gemini key, test on phone, rehearse, MVP due	Today

Roadmap after the MVP

1. **Accounts & push (1-2 weeks):** Supabase sign-in and sync, Web Push when the app is closed, nightly re-plan job.
2. **Smarter planning (2-3 weeks):** Google Calendar sync, syllabus/brief upload, learning real vs estimated time, energy model.
3. **Accountability layer (2 weeks):** pair with a friend, shared streaks, weekly report.
4. **Native & teams (2027):** Android/iOS apps with widgets, coach and campus editions, goal templates.